

JANS FAHREL ADIPUTRA MANURUNG

Computer Technology Student • Backend & IoT Enthusiast

North Sumatra, Indonesia | (+62) 87-73060-4407 | jansfahrel@gmail.com

LinkedIn: [jansfahrel](#) | GitHub: [jansfahrel](#)

SUMMARY

Highly motivated Diploma (D3) student in Computer Technology at Institut Teknologi Del, with a strong interest in software development and the Internet of Things (IoT). Experienced in backend development, database management, server-side programming, and IoT integration. Enjoys bridging hardware and software to build secure, scalable, and efficient digital solutions. Career objective: to become a proficient Backend Developer.

EDUCATION

Institut Teknologi Del

Associate Degree (D3) in Computer Technology

PROJECTS

Toba and Samosir MSME Website | Backend Developer

Developed a web-based platform to digitalize and promote local MSMEs in the Toba and Samosir regions, providing a digital storefront to expand market reach.

Technologies: Laravel, MySQL

- Designed and built the relational database architecture using MySQL to manage product and user data efficiently.
- Developed core backend logic and assisted in building website features using Laravel for seamless integration and functionality.

E-Tugas (E-Assignment) Application | Full-Stack Developer

Developed a web-based assignment management application from scratch to streamline task tracking and submissions.

Technologies: Laravel, MySQL, Docker, Ubuntu

- Led end-to-end development, including frontend interface and backend logic using Laravel.
- Designed the MySQL database architecture and used Docker to deploy a multi-node database setup across two laptops, providing data redundancy and backup.
- Configured an Ubuntu server to host the application over a local network for access from mobile devices.

GitHub: [jansfahrel/aplikasi_e-tugas](#)

Smart Room Security System | IoT Developer

Developed an IoT-based smart security camera system that detects unauthorized motion and sends real-time photo alerts.

Technologies: ESP32-CAM, PIR Sensor, C/C++ (Arduino IDE), Telegram Bot API, Fritzing, SolidWorks

- Integrated an ESP32-CAM with a PIR sensor to automatically capture images when motion is detected.
- Programmed Telegram Bot API integration for instant photo transmission and real-time security alerts.
- Designed the hardware wiring layout in Fritzing and modeled the physical casing in SolidWorks.

GitHub: [GitHub - jansfahrel/esp32-cam-room-security- · GitHub](#)

IoT Soil Nutrient Monitoring System | IoT Developer

Developed an IoT-based agricultural device to measure soil nutrient levels and transmit collected data to a web platform for real-time monitoring and storage.

Technologies: Microcontroller (ESP32/ESP8266), Soil Sensors, RESTful API, Web Server, Database

- Assembled and wired hardware components, integrating soil sensors with the microcontroller.
- Configured API integration between the IoT hardware and web server.
- Programmed real-time transmission of soil measurements to the backend for storage and user access.

GitHub: [jansfahrel/Perangkat-Pengukur-Unsur-Hara-Tanah-Tugas-Akhir](#)

TECHNICAL SKILLS

Backend & Web Development: Laravel, Python, RESTful API

Database Management: MySQL, MongoDB

IoT & Embedded Systems: C/C++, ESP32, Sensor Integration

Hardware Design: Fritzing, SolidWorks

Infrastructure & Deployment: Docker, Ubuntu (Linux), Vercel

Soft Skills: Complex Problem Solving, Analytical Thinking, Team Collaboration, Adaptability & Fast Learning

CERTIFICATIONS

Cloud Basics and Gen AI on AWS • Programming Logic 101 • JavaScript Programming Fundamentals • Structured Query Language (SQL) Basics • Data Classification & Summarization • Android UI Development • Java Programming • Data Science Basics • Mobile Application Development

ORGANIZATIONAL EXPERIENCE

Program Cinta Almamater (PCA) Committee — Sports Division Member

- Educated and oriented new students about the campus environment, core values, and culture of Institut Teknologi Del.

Abang Asuh (Dormitory Mentorship Program) — Mentor

- Guided and mentored junior students in adapting to dormitory life, including daily routines, regulations, and community living.